

Decision and Risk Analysis

Game Theory

DAEN 427/ISEN 427 Fall 2025
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Why Game Theory Exists

Coordination and Nash Equilibrium

The Prisoners' Dilemma

Mixed Strategies

Refinements and Sequential Games

Cooperation and Culture

From Incomplete Information to Bayesian Games

Wrapping Up

From Parametric to Strategic Decisions

Parametric decisions: The outcome depends on your action and nature's state.

"Should I bring an umbrella?"

It depends on the weather, not what others do.

Strategic decisions: The outcome depends on your action *and* others' actions.

"Should I bring an umbrella when there's only one dry bus seat left?"

Strategic interaction means anticipating others' moves.
You're not just solving a problem, you're solving people.

- **Game theory** is the study of *mathematical models of strategic interactions* between rational decision-makers.
- Applications in many fields: economics, logic, computer science, behavioral science, biology.
- Modern game theory began with the work of John von Neumann and Oskar Morgenstern in their 1944 book *Theory of Games and Economic Behavior*.
- John Nash introduced the Nash equilibrium in the 1950s, extending the theory to non-zero-sum games.



von Neumann

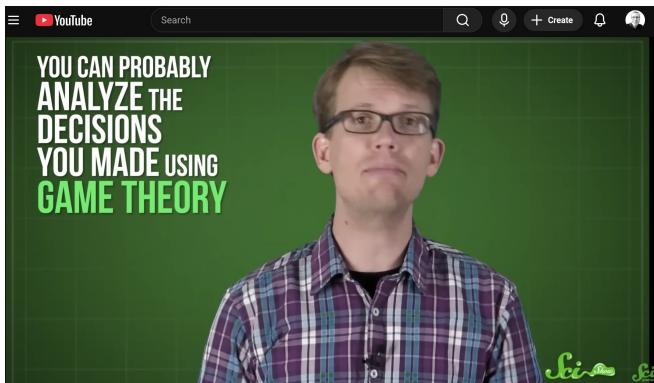


Nash



A Beautiful Mind

A nice video (for your reference)

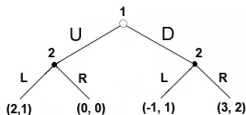


Key Types of Games

- **Cooperative vs Non-cooperative**
 - Cooperative: players can form binding agreements.
 - Non-cooperative: players act individually.
- **Symmetric vs Asymmetric**
 - Symmetric: players have identical strategy sets and payoffs only depend on strategies, not on player identity.
- **Zero-sum vs Non-zero-sum**
 - Zero-sum: one player's gain = another's loss.
 - Non-zero-sum: players may all gain or lose.
- **Simultaneous vs Sequential**
 - Simultaneous: players move without knowing other's choice.
 - Sequential: players move in turn, possibly observing earlier moves.

Representation of Games

- **Normal form (strategic form):** payoff matrix showing each player's strategies & payoffs.
- **Extensive form:** decision tree showing sequential moves, information sets.
- **Characteristic function form:** used in cooperative games, describing value of each coalition.



Extensive Form

	(L,L)	(L,R)	(R,L)	(R,R)
U	2,1	2,1	0,0	0,0
D	-1,1	3,2	-1,1	3,2

Strategic Form

$$v(c) = \begin{cases} 80, & \text{if } c = \{A\} \\ 56, & \text{if } c = \{B\} \\ 70, & \text{if } c = \{C\} \\ 80, & \text{if } c = \{A, B\} \\ 85, & \text{if } c = \{A, C\} \\ 72, & \text{if } c = \{B, C\} \\ 90, & \text{if } c = \{A, B, C\} \end{cases}$$

Function Form

Applications

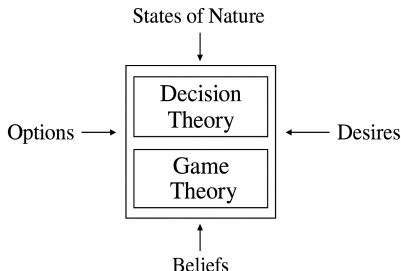
- **Economics:** oligopoly, auctions, bargaining.
- **Biology:** evolutionary strategies, competition.
- **Political science:** voting, war/peace strategies.
- **Social sciences:** negotiation, networks, behavioral.
- **Computer science / AI:** multi-agent systems, system design.

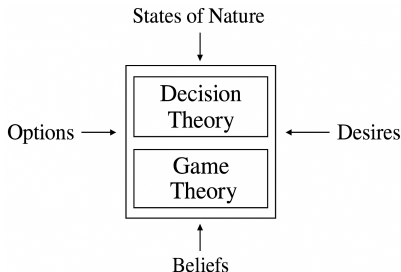


Although not always perceived as such, the theory is actually **multidisciplinary**.

Well-Known Examples

- **The Prisoner's Dilemma:** Illustrates why two rational individuals might not cooperate, even if it appears that it is in their best interest to do so.
- **The Battle of the Sexes:** Two players prefer to coordinate but have different preferences for coordination outcomes.
- **The Ultimatum Game:** A proposer offers a split of a sum; the responder accepts or rejects.





- Helps in understanding strategic behavior when outcomes depend on others' actions.
- Provides rigorous tools for modeling interaction in a wide range of fields.
- Highlights how rational decision-making can lead to suboptimal outcomes (e.g., Prisoner's Dilemma).
- Supports design of mechanisms, policies, or strategies where multiple decision-makers interact.

The Makeup Exam (A Classic Lie Gone Wrong)

Two students skip their exam and claim a flat tire.

Professor gives a makeup test.

Question 1 (5 points): Easy stats.

Question 2 (95 points): "Which tire?"

- Both write the same tire → both get A's.
- Different tires → both fail miserably.

This is a **coordination game**, the best outcome depends on matching the other player's answer. *Even lies require equilibrium.*



The Makeup Exam (A Classic Lie Gone Wrong)

Payoff Matrix:

	FL	FR	RL	RR
FL	A, A	F, F	F, F	F, F
FR	F, F	A, A	F, F	F, F
RL	F, F	F, F	A, A	F, F
RR	F, F	F, F	F, F	A, A

- Same tire $\rightarrow$ Both get A's.
- Different tires $\rightarrow$ Both fail.
- **Four Nash equilibria:** one for each tire.

The Prisoners' Dilemma

Two suspects. Two choices: Cooperate (C) or Defect (D).

Payoff Matrix (in years in jail):

	C	D
C	$(-2, -2)$	$(-20, 0)$
D	$(0, -20)$	$(-10, -10)$

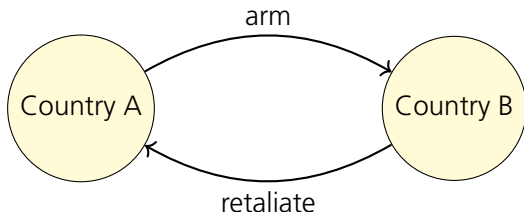
Nash Equilibrium: Both defect. Because defection strictly dominates cooperation.

Lesson: Rational people can end up in irrational situations. Just like every group project ever.



In the Real World

- **Arms races:** Each country wants safety, ends up broke.
- **Overfishing:** Everyone wants "just one more fish."
- **Pollution:** Everyone drives, nobody breathes.
- **Corporate meetings:** Everyone talks, nothing gets done.



Individual rationality doesn't guarantee collective happiness.
Sometimes the "smart" move ruins the world (or the weekend).

Repeated Games: Hope for Humanity?

If the game repeats indefinitely, cooperation can emerge.
Why? Because players can *punish* defectors in future rounds.

Example: “You didn’t do the dishes yesterday—guess who’s sleeping on the couch tonight?”

But if there’s a last round, backward induction kills cooperation:

- Last round $\rightarrow$ no reason to cooperate.
- Therefore, second to last $\rightarrow$ no reason either.

Lesson: The only way to cooperate forever is to never end the game.

Nash Equilibrium (“No Regrets” Rule)

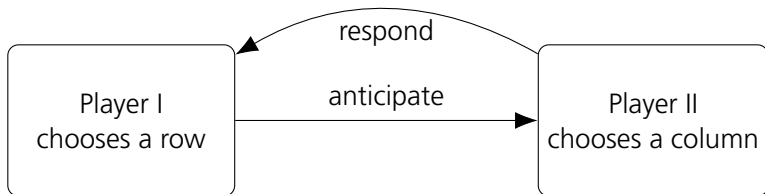
- A **Nash equilibrium** is a concept in game theory where no player can improve their outcome by changing their own strategy, assuming all other players keep their strategies unchanged.
- In simpler terms: everyone is doing the best they can given what others are doing. **No one has an incentive to deviate.**

Example: In the Prisoner’s Dilemma, both players choosing to “confess” is a Nash equilibrium — neither can get a better outcome by unilaterally switching to “stay silent.”

So, a Nash equilibrium represents a stable state where all players’ decisions are mutually consistent.

Nash Equilibrium ("No Regrets" Rule)

Definition: Each player's strategy is the best response to others'. No one can improve by unilaterally changing.



Example: In the flat tire, both write "Front Left."
Neither can improve → equilibrium.

Nash Equilibrium (The Logic of "No Regrets")

Definition: A Nash equilibrium is a strategy profile where each player's choice is the best response to the others' choices.

In plain English: "Given what you're doing, I can't do better by changing."

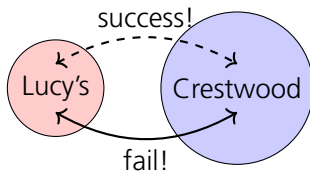
Example: In the flat tire, if both write "Front Left," neither gains by switching. $\Rightarrow$ Each is doing the best they can given the other's (bad) lie.

Moral: Equilibrium $\neq$ best outcome. It just means nobody can complain *without being a hypocrite*.

Coffee Shop Coordination

You and your friend forgot to pick which café, Lucy's or Crestwood.

	Lucy's	Crestwood
Lucy's	(1,1)	(0,0)
Crestwood	(0,0)	(1,1)



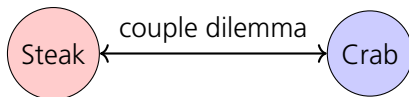
Two pure Nash equilibria:

- (Lucy's, Lucy's)
- (Crestwood, Crestwood)

Moral: Coordination is easy if you have telepathy or text messages (unless you're trying to text each other from the subway).

The Battle of the Sexes (aka Netflix Negotiations)

A couple wants to dine together but have different preferences.



		Steak House	Crab Shack
Payoffs:	Steak House	(2,1)	(0,0)
	Crab Shack	(0,0)	(1,2)

Equilibria:

- (Steak House, Steak House)
- (Crab Shack, Crab Shack)

Analogy: Like when one wants to watch "John Wick" and the other "Bridgerton." Either way, someone's pretending to enjoy it.

Lesson: Rational harmony sometimes means pretending to enjoy the other's choice.

When Pure Strategies Fail

Sometimes there's no equilibrium in pure strategies. Enter the **mixed strategy**.

Example: Matching Pennies (or Coffee Shops Redux)

You want to avoid your ex; your ex wants to find you.

No pure equilibrium, but a mixed one exists: both randomize 50/50.

Translation: If you can't predict them, confuse them. (Also known as "dating.")

Battle of the Sexes in Mixed Strategies

In mixed equilibrium:

- Player I goes to Steak House with probability $p = 2/3$.
- Player II goes to Crab Shack with probability $q = 1/3$.

Both are indifferent between choices.

Takeaway: You can't win every time—but you can confuse your partner enough that no one knows who's losing.

Spousonomics (Game Theory at Home)

Book premise: “Economics is the route to marital bliss.”

Example: Randomize chores.

- Sometimes do the laundry.
- Sometimes don't.
- Keep your partner guessing.

Result: Higher household efficiency. Possibly higher divorce risk.
Mixed strategies: not just for poker—also for surviving marriage.

Trembling-Hand Perfection

Even rational players slip up (literally tremble). A strategy is **trembling-hand perfect** if it still works when others might make tiny mistakes ($\epsilon > 0$).

Example: You “accidentally” like your ex’s 2017 photo. If your strategy survives that tremble, it’s perfect.

Key: Robust equilibria survive human clumsiness.

Subgame Perfection (Planning Every Move)

Sequential games have multiple stages. You must think ahead and reason backward (**backward induction**).

Example: Parent and child:

- Child: "If I cry, I get candy."
- Parent: "If I give candy now, I'll regret it later."

Backward induction says: think from the end. **Moral:** Rational parenting = consistent punishment and lots of snacks.

MAD – Mutually Assured Destruction

Nuclear strategy: “If you bomb me, we all die.”

It's a Nash equilibrium, but not **subgame perfect** because the threat isn't credible.

Solution: Dr. Strangelove's doomsday machine, automatic retaliation!

Lesson: To make a threat credible, automate it ... or just build something so terrifying that no one calls your bluff.

The Stag Hunt – Trust and Coordination

Two hunters can chase a deer (big reward) or hunt hares (small reward).

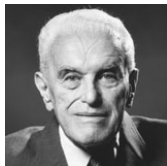
	Deer	Hare
Deer	(3,3)	(0,1)
Hare	(1,0)	(1,1)

Two equilibria:

- Both cooperate for deer → high reward.
- Both settle for hares → safe but boring.

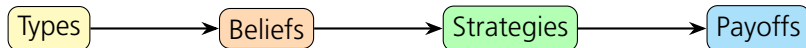
Moral: Societies thrive when people trust others to hunt the deer.

From Incomplete Information to Bayesian Games



- Introduced by John C. Harsanyi (1967–68).
- A **Bayesian game** models strategic interaction with private information.
- Each player has a **type** that represents hidden knowledge.
- Players have **beliefs** (probability distributions) about others' types.

Flow Diagram

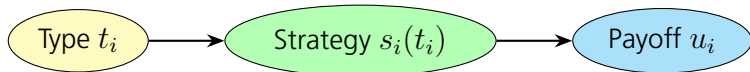


Structure of a Bayesian Game

A Bayesian game consists of:

- Players $i = 1, 2, \dots, n$
- Action sets A_i
- Type sets T_i
- Common prior $p(t_1, t_2, \dots, t_n)$
- Payoffs $u_i(t_1, \dots, t_n, a_1, \dots, a_n)$

Dependency Schematic



Strategies and Beliefs

- Each player chooses a strategy function $s_i : T_i \rightarrow A_i$ that maps types to actions.
- Players maximize expected utility based on their beliefs.

Example: Auction

- Each bidder's valuation (type) is private.
- Strategy: bid as a function of valuation.

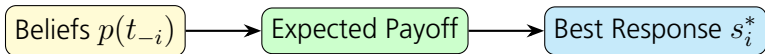
Bayesian Nash Equilibrium (BNE)

A **Bayesian Nash Equilibrium** is a strategy profile $(s_1^*, \dots, s_n^*)$ such that for every player i and every type t_i ,

$$s_i^*(t_i) \in \arg \max_{a_i \in A_i} \mathbb{E}_{t_{-i}}[u_i(a_i, s_{-i}^*(t_{-i}), t_i, t_{-i})].$$

- Each player's strategy maximizes expected payoff given beliefs and others' strategies.
- Generalizes Nash equilibrium to settings with private information.

Optimization Flowchart



Homework Assignment

Due: Thu Nov 20 9:35 AM

Read → answer the **blue** questions!

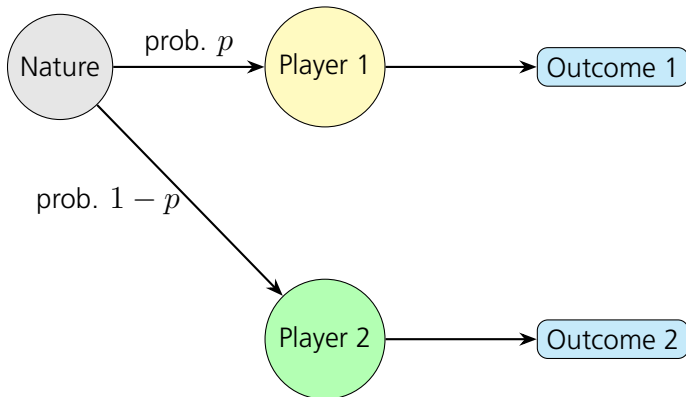
Use bullet points or full sentences.

Stochastic Outcomes

- Many games include **random events** that affect payoffs (e.g., dice rolls, market shocks).
- These are represented using a random variable or a special **“Nature” player**.
- A stochastic game may involve:
 - Randomly determined states or transitions.
 - Probabilistic payoffs based on chance.
 - Belief updates when new information is revealed.
- Players maximize their **expected utility** across all possible outcomes.

Stochastic Outcomes

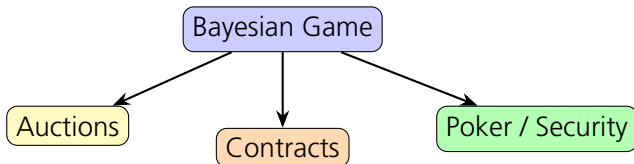
Nature's Move Diagram



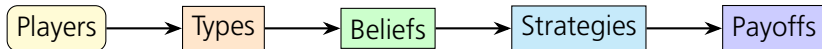
Applications of Bayesian Games

- **Auctions:** Players bid based on private valuations.
- **Contract theory:** Principal–agent problems with hidden information.
- **Adversarial settings:** Poker, security resource allocation.
- **Market competition:** Firms with private cost data.

Concept Map



- Game theory analyzes strategic interactions among rational players.
- Bayesian games model incomplete information using types and beliefs.
- Players choose strategies based on expected utilities.
- The key equilibrium concept: **Bayesian Nash Equilibrium**.



The World as a Game

Game theory teaches:

- People don't act alone, they anticipate others.
- Nash equilibria = stable, but not necessarily happy.
- Cooperation is fragile but possible (or, not impossible :)
Remember: Every group chat, meeting, and marriage is secretly a repeated Prisoners' Dilemma.

In real life:

- Rational $\neq$ sensible.
- Mixed strategies are underrated.
- And every social interaction is a low-stakes prisoners' dilemma.

"Life is a series of coordination failures disguised as social events."

Main takeaway:

- Know the game.
- Anticipate others' moves.
- And if you lie about a flat tire, agree on which one.

One More Thing ...

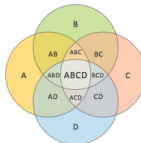
- 7 sets Venn Diagram
- Venn Diagrams



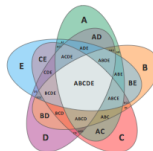
2-set Venn diagram



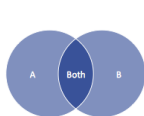
3-set Venn diagram



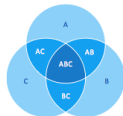
4-set Venn diagram



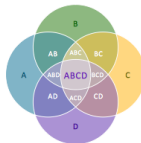
5-set Venn diagram



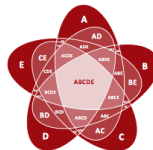
2-set Venn diagram



3-set Venn diagram



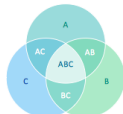
4-set Venn diagram



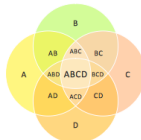
5-set Venn diagram



2-set Venn diagram



3-set Venn diagram



4-set Venn diagram



5-set Venn diagram

Evolutionary dynamics of any multiplayer game on regular graphs

